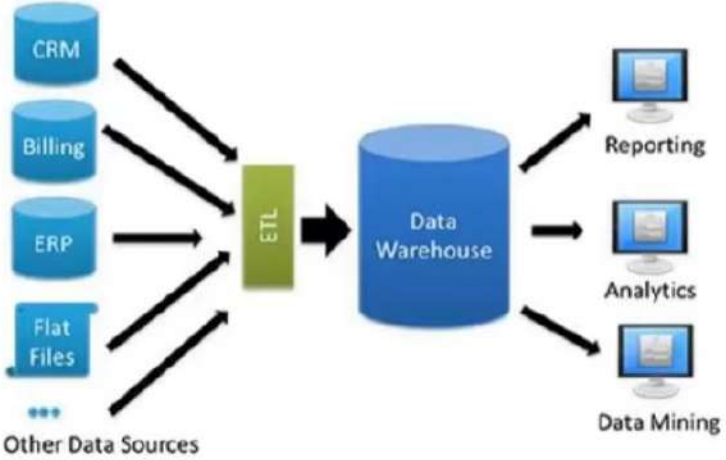
What is HIVE ?

- Hive is a data warehouse infrastructure that is built on top of Hadoop
- Hive provides a mechanism to project structure onto the data and query the data using SQL-like language called HiveQL
- Hive uses Map Reduce and HDFS for Processing and storage/ retrieval of data



HIVE Advantages

The advantages of using Hive are:

- It can be used as an ETL.
- Provides capability of querying and analysis.
- Can handle large data sets.
- SQL(filters, joins, group by) on top of Map and Reduce.

Where not to use HIVE

Hive should not be used:

- If the data does not cross GB's.
- If we don't find schema or bringing in schema is difficult or not possible on the data at hand.
- If we need response in seconds and for low latency applications.
- If RDBMS can solve, don't invest time in Hive.

HIVE Features

Similarity with SQL

- Hive is very similar to RDBMS SQL like queries
- HiveQL is based on SQL-92 framework and it is safe to say that its functions as SQL for Hadoop DFS.

Difference from SQL

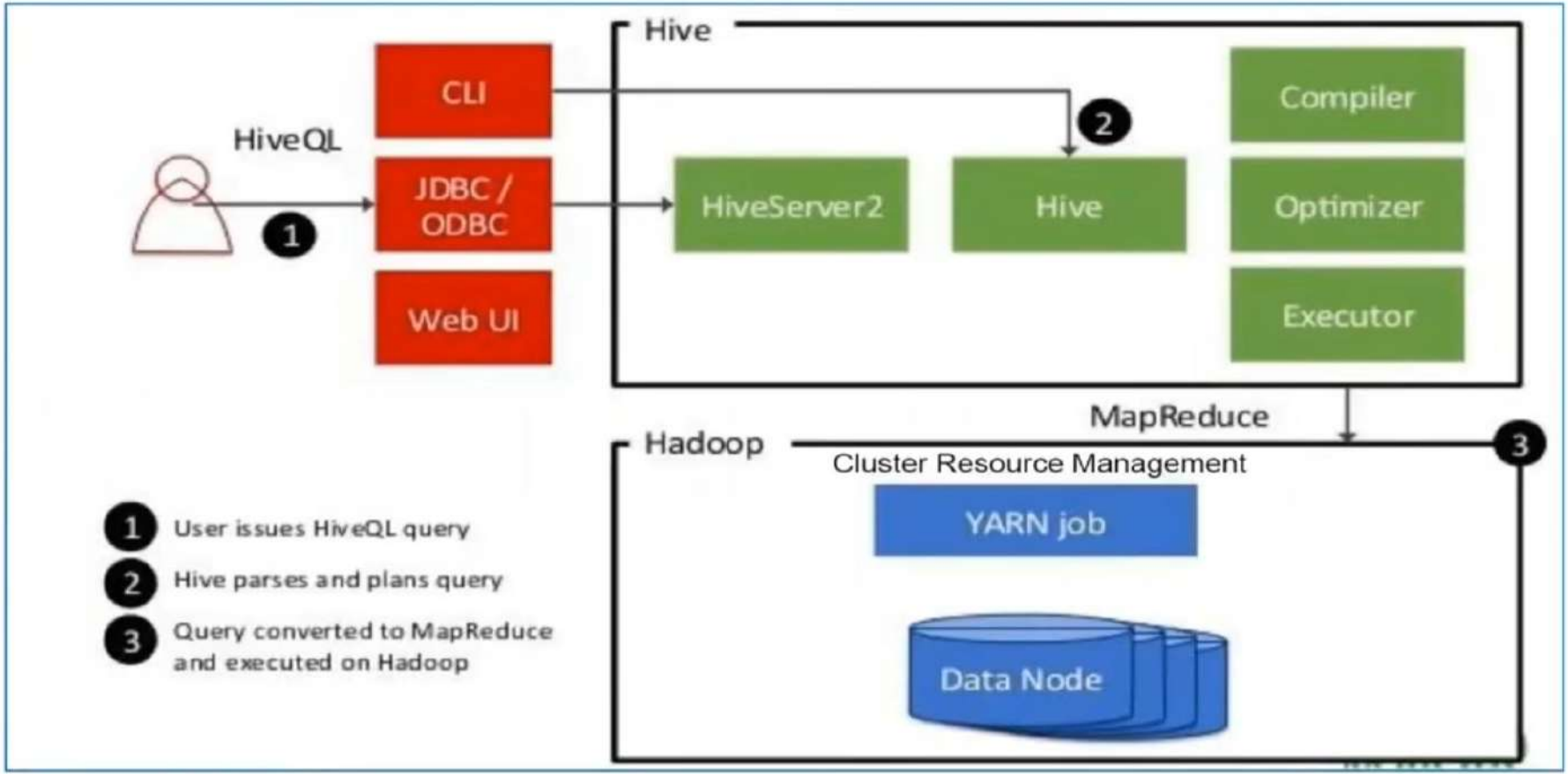
- The major difference is that a Hive query executes on a Hadoop infrastructure rather than a traditional database.
- This allows Hive to scale to handle huge data sets - data sets so large that high-end, expensive, traditional databases would fail.
- The internal execution of a Hive query is via a series of automatically generated Map Reduce jobs

HIVE Vs MapReduce

All Hive queries will be converted to Map Reduce Job for execution, so why can't we write Map Reduce our selves?

- Understanding the internals of Hadoop Framework is must to write Map Reduce Jobs.
- Engineer having SQL knowledge can quickly write Hive Scripts and get the results.

HIVE Architecture

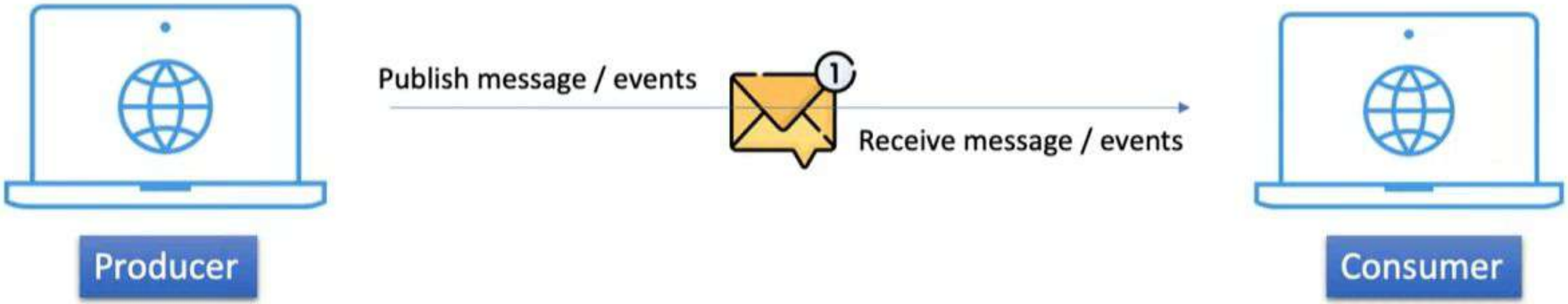


Kafka

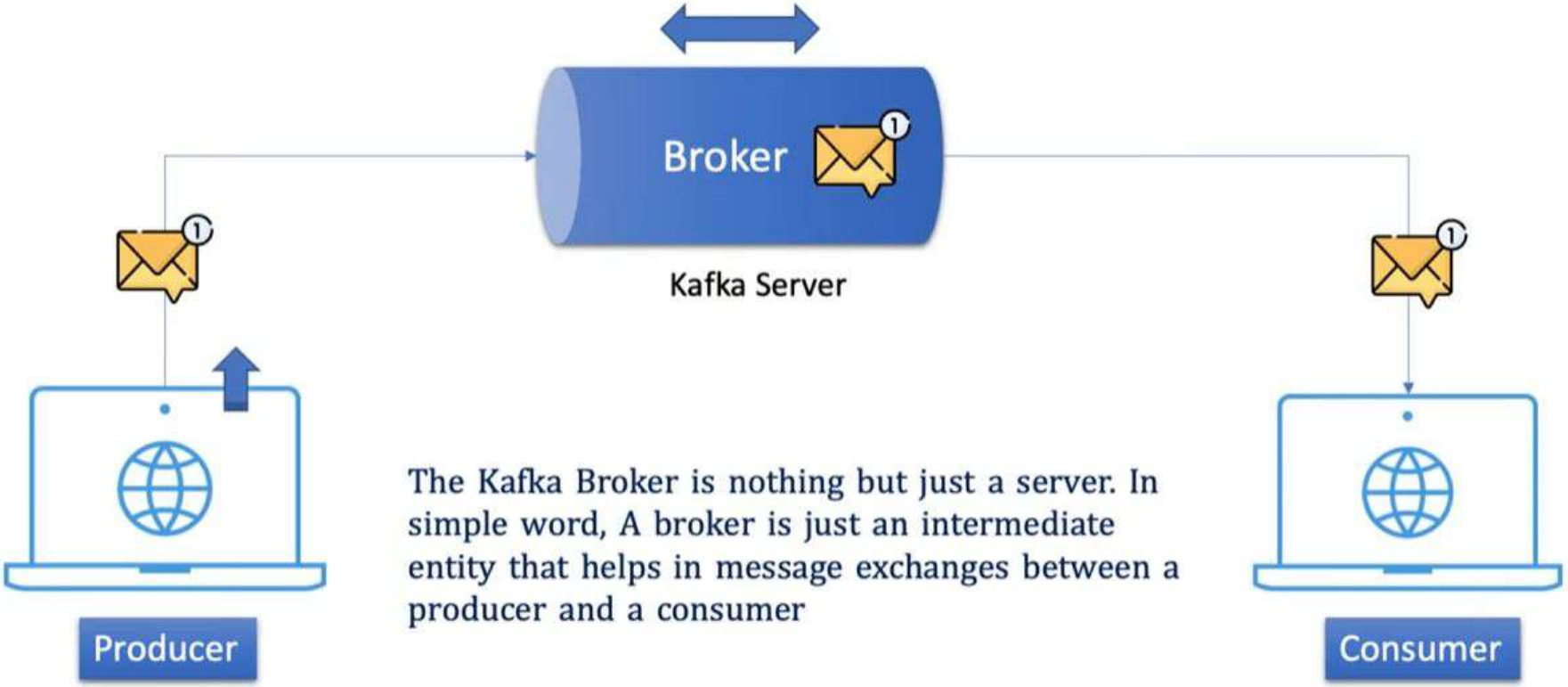
Kafka Component

- Producer
- Consumer
- Broker
- Cluster
- Topic
- Partitions
- Offset
- Consumer Groups
- Zookeeper

Kafka

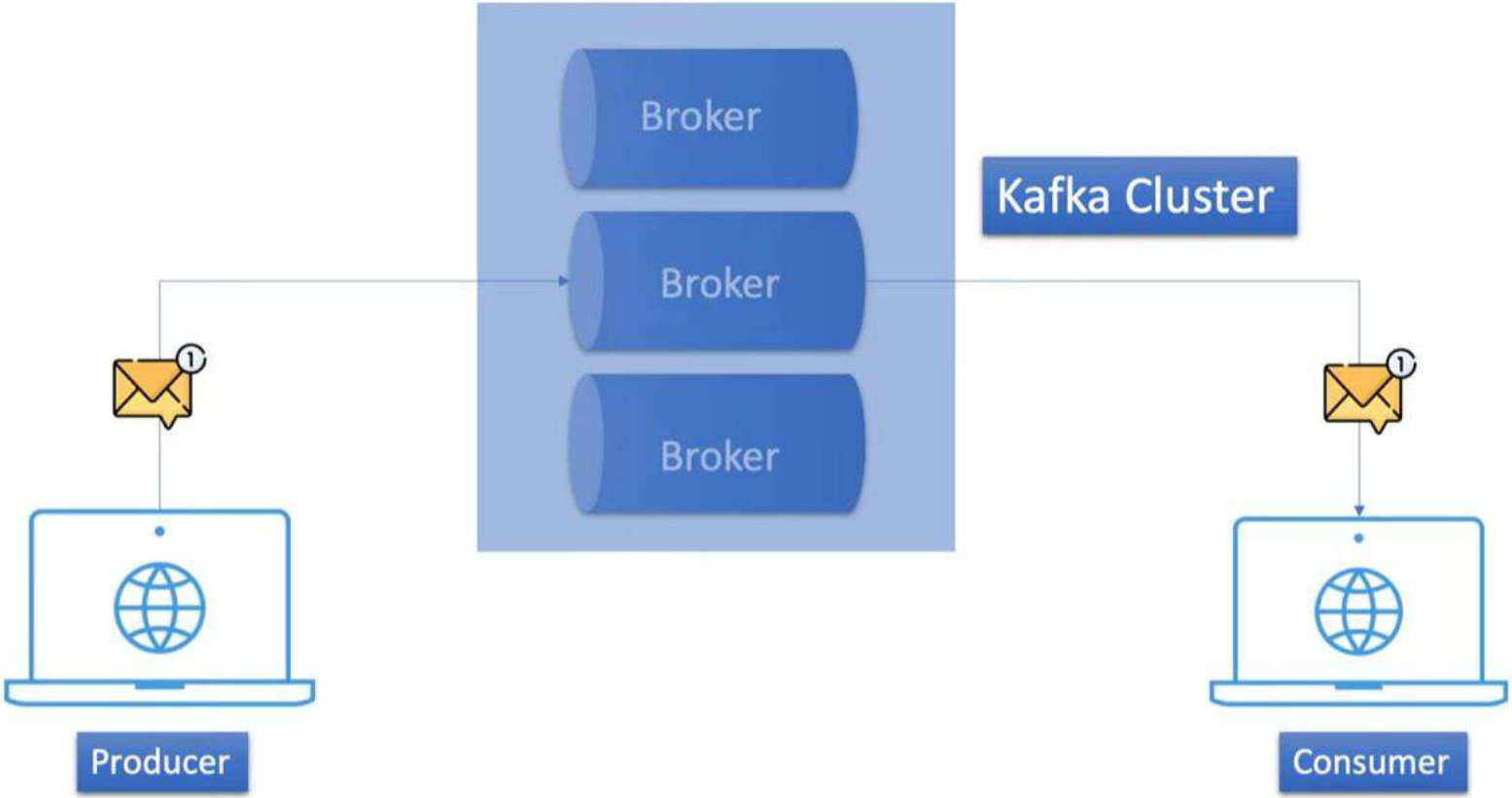

Publish message / events
1
Receive message / events
Producer
Consumer

Kafka Component

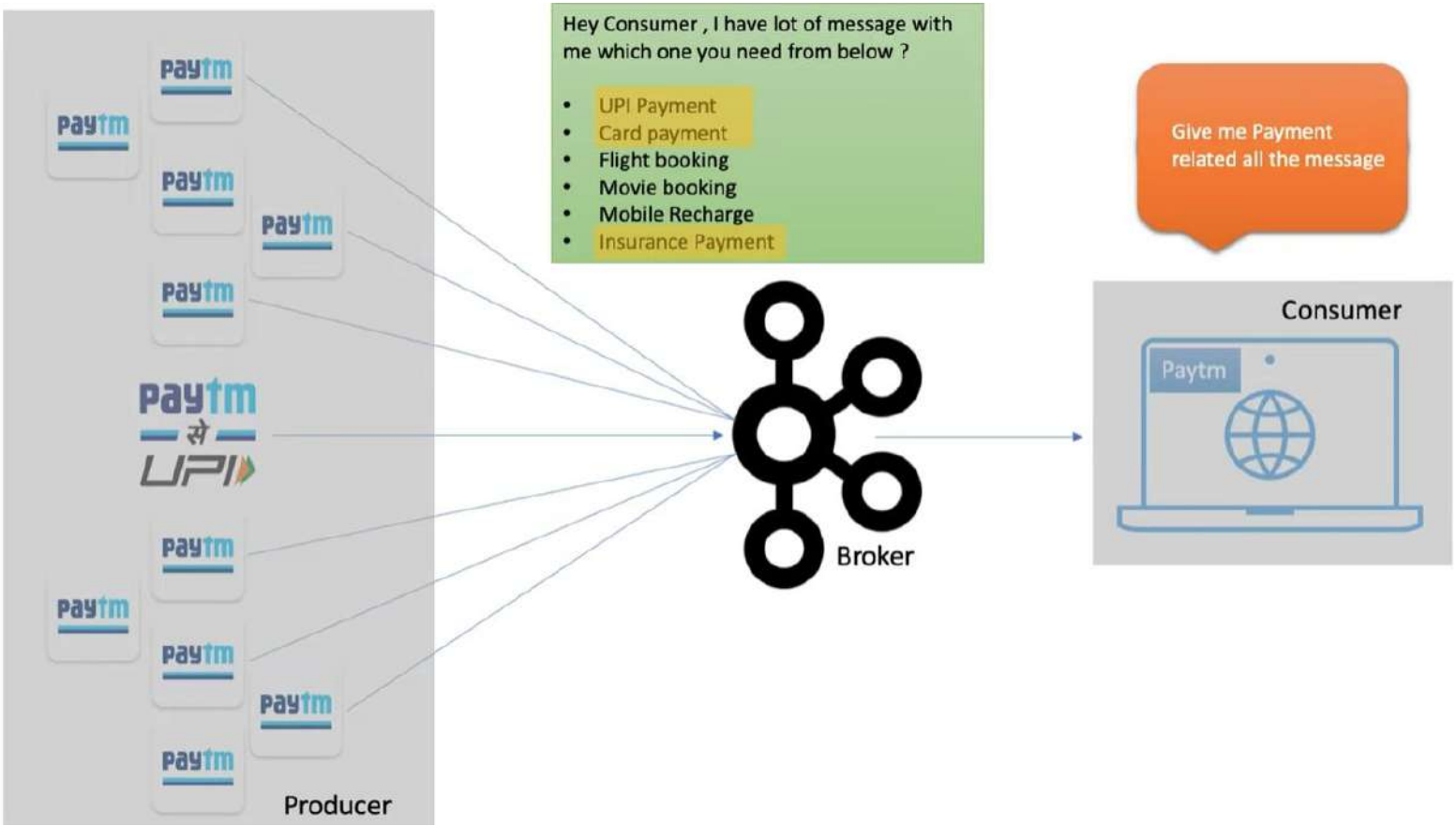


The Kafka Broker is nothing but just a server. In simple word, A broker is just an intermediate entity that helps in message exchanges between a producer and a consumer

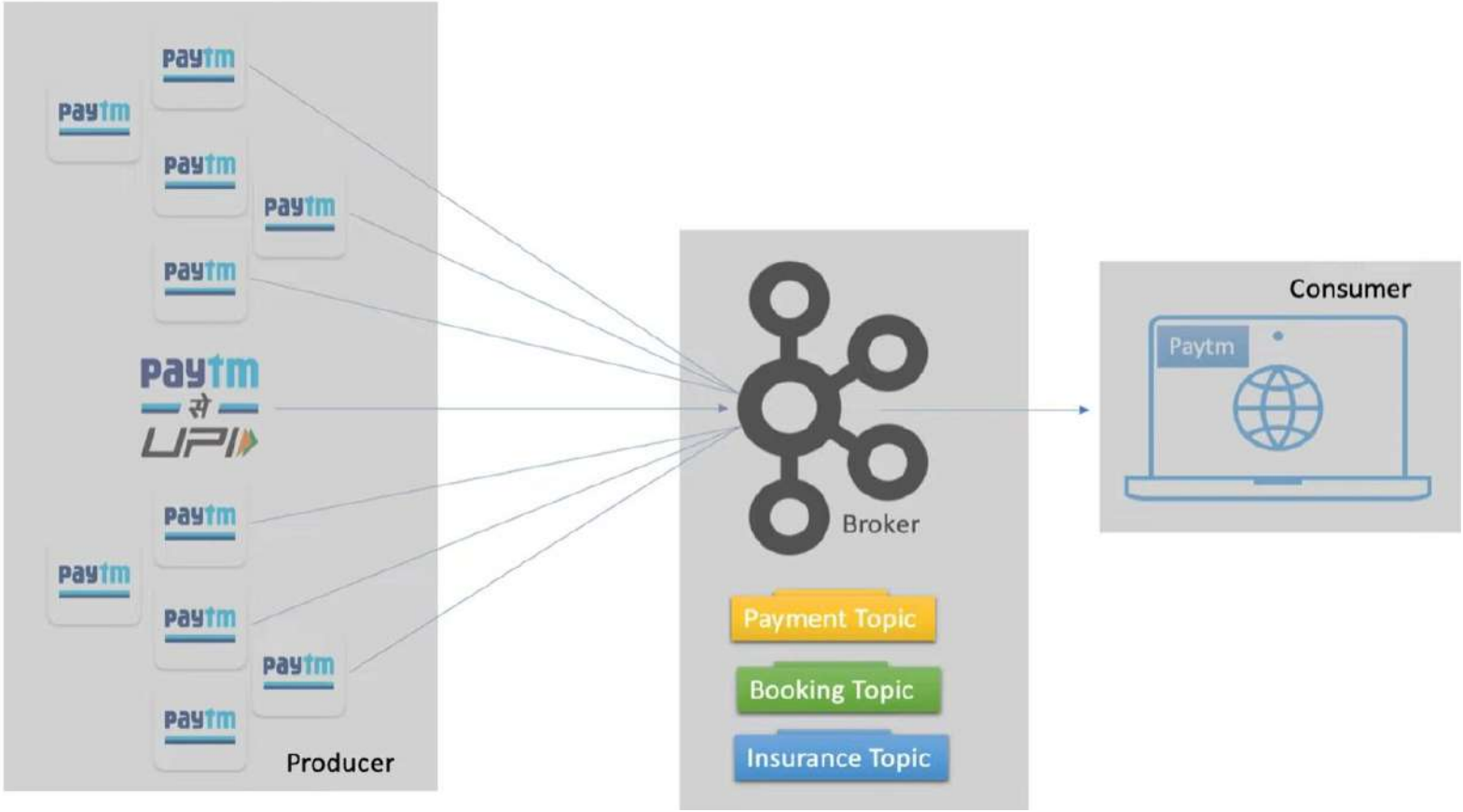
Kafka Cluster



Topic (Why it is inside the broker?)



Topic (Why it is inside the broker?)



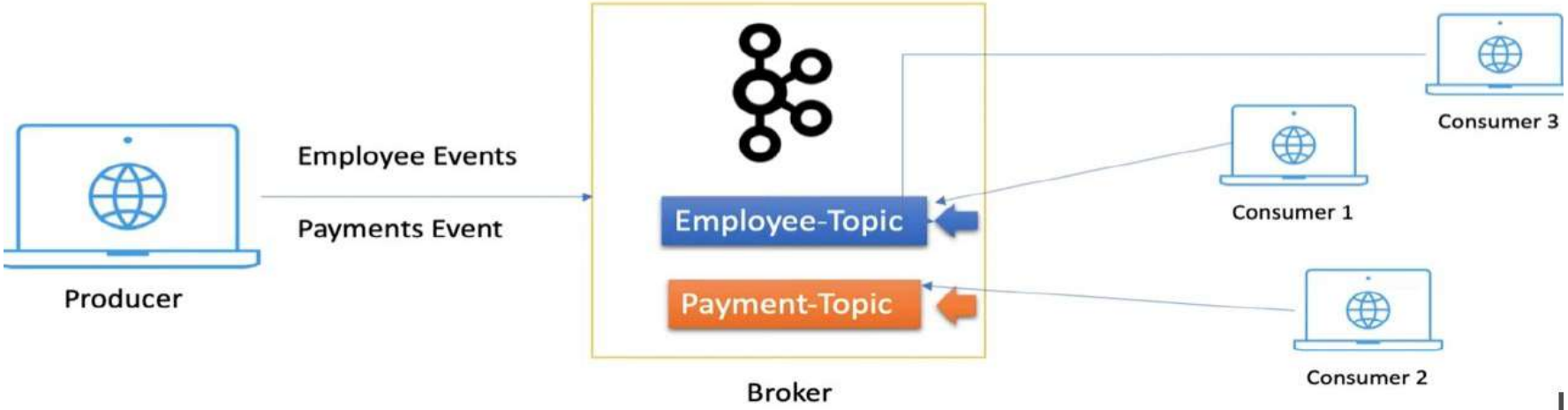
Topic (Act as Databse Table)

EMPLOYEE_TABLE

ID	NAME	DEPT	SALARY
1	john	IT	50000
2	sam	Bank	80000
3	joe	Admin	20000

PAYMENT_TABLE

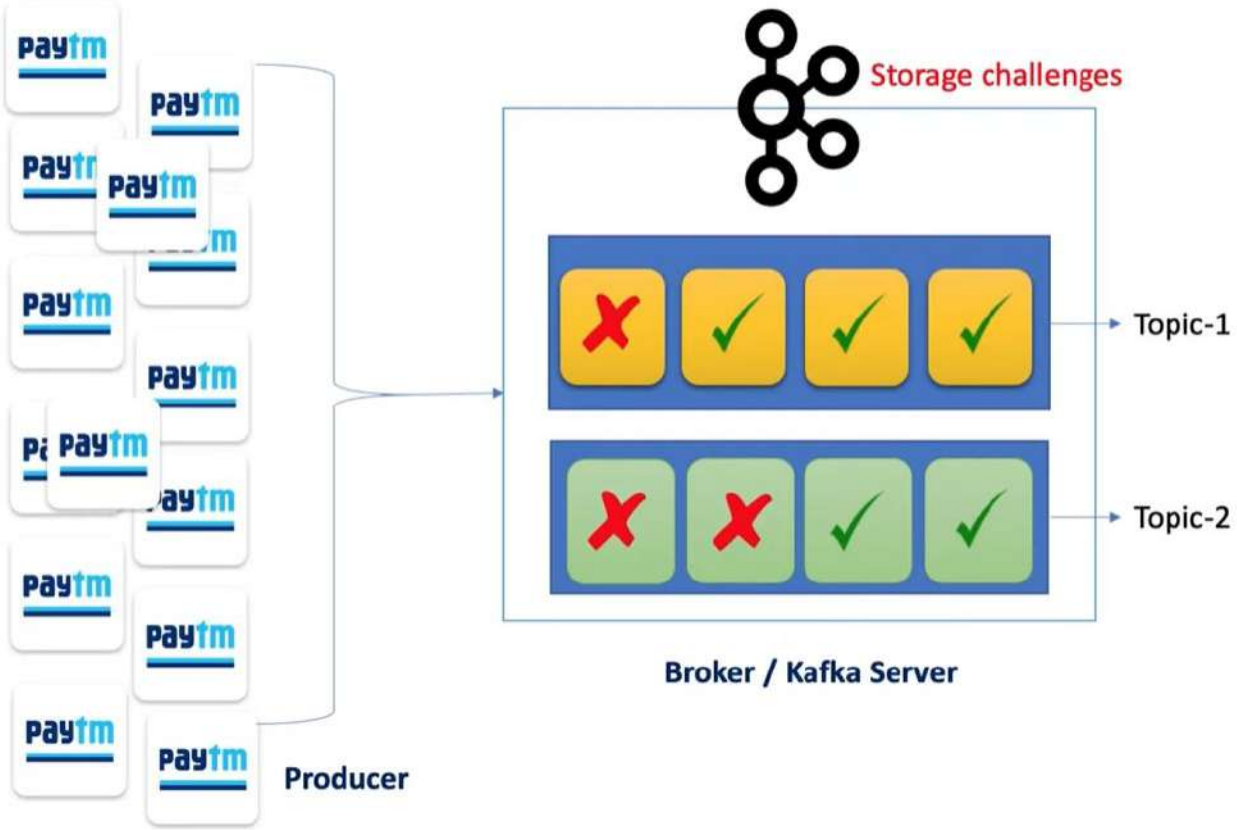
PAYMENT_ID	CNAME	PAY MODE	SRC AC	DEST AC	AMOUNT
df364bdjfb	Basant	CREDIT	HDFC12	SBI239837	100000
bcsr72386d	Santosh	DEBIT	ICICI893	HDFC3626	890012
dfv834tjsd8	Krishna	CHEQUE	SBI2367	IDBI23726	23273582



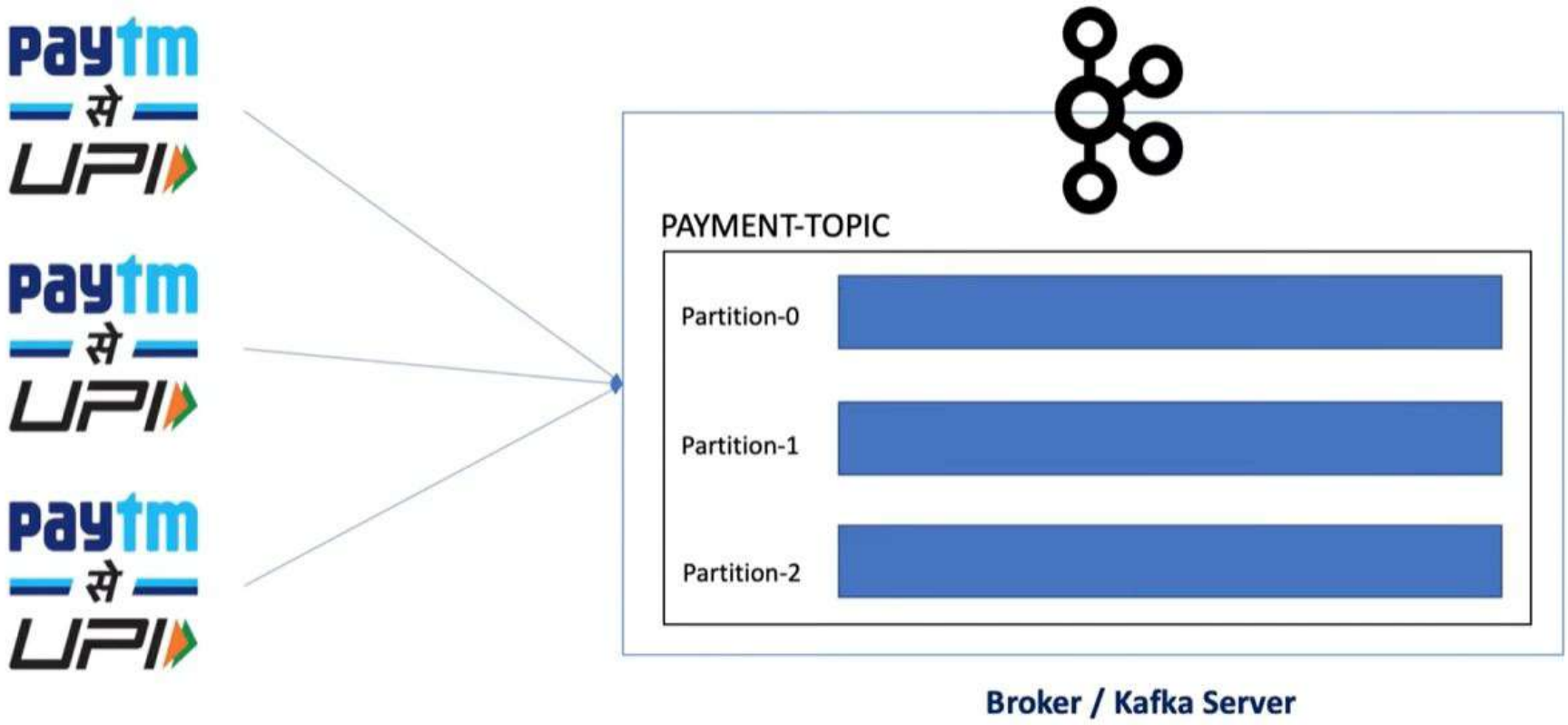
Topic

It specifies the category of the message or the classification of the message. Listeners can then just respond to the messages that belong to the topics they are listening on.

Partitions

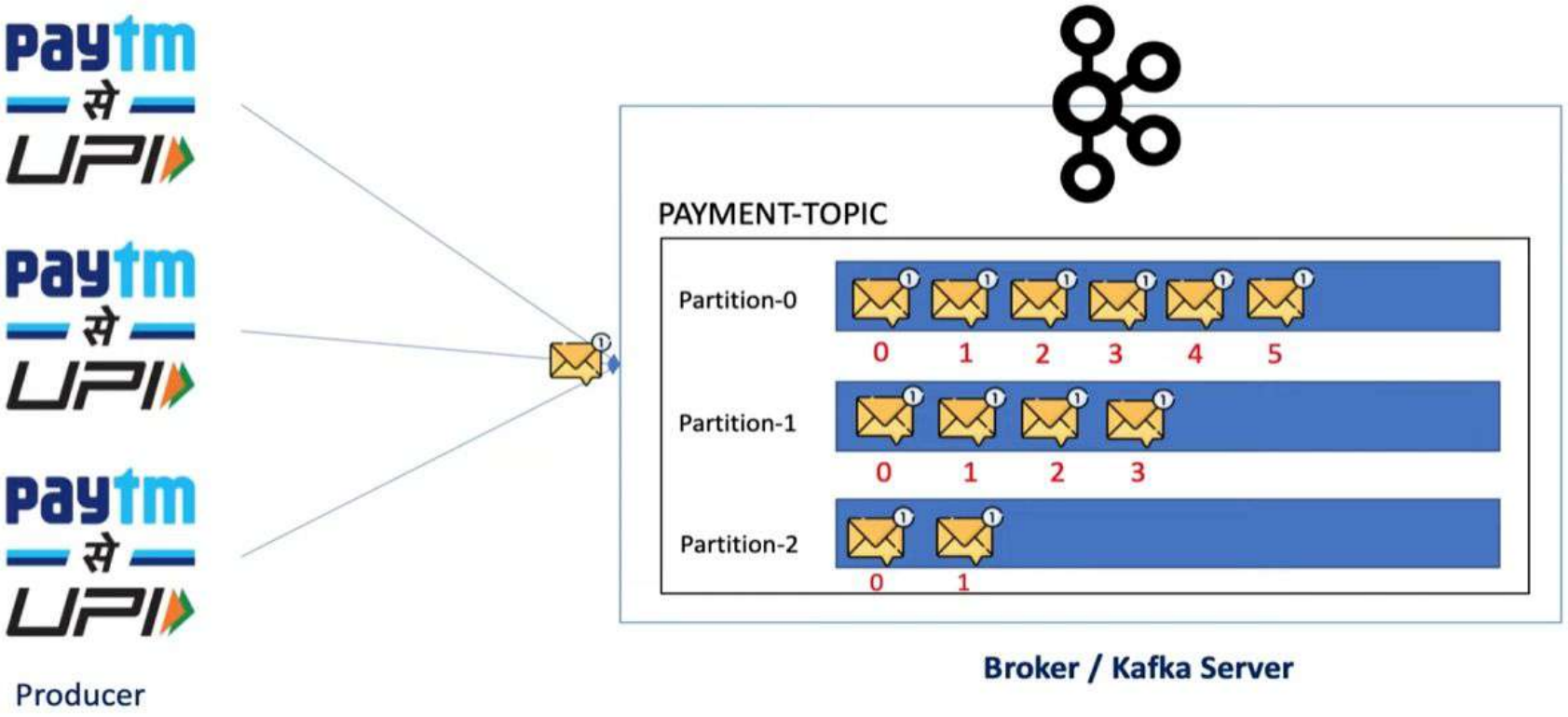

Storage challenges
Topic-1
Topic-2
Broker / Kafka Server
Producer

Partitions

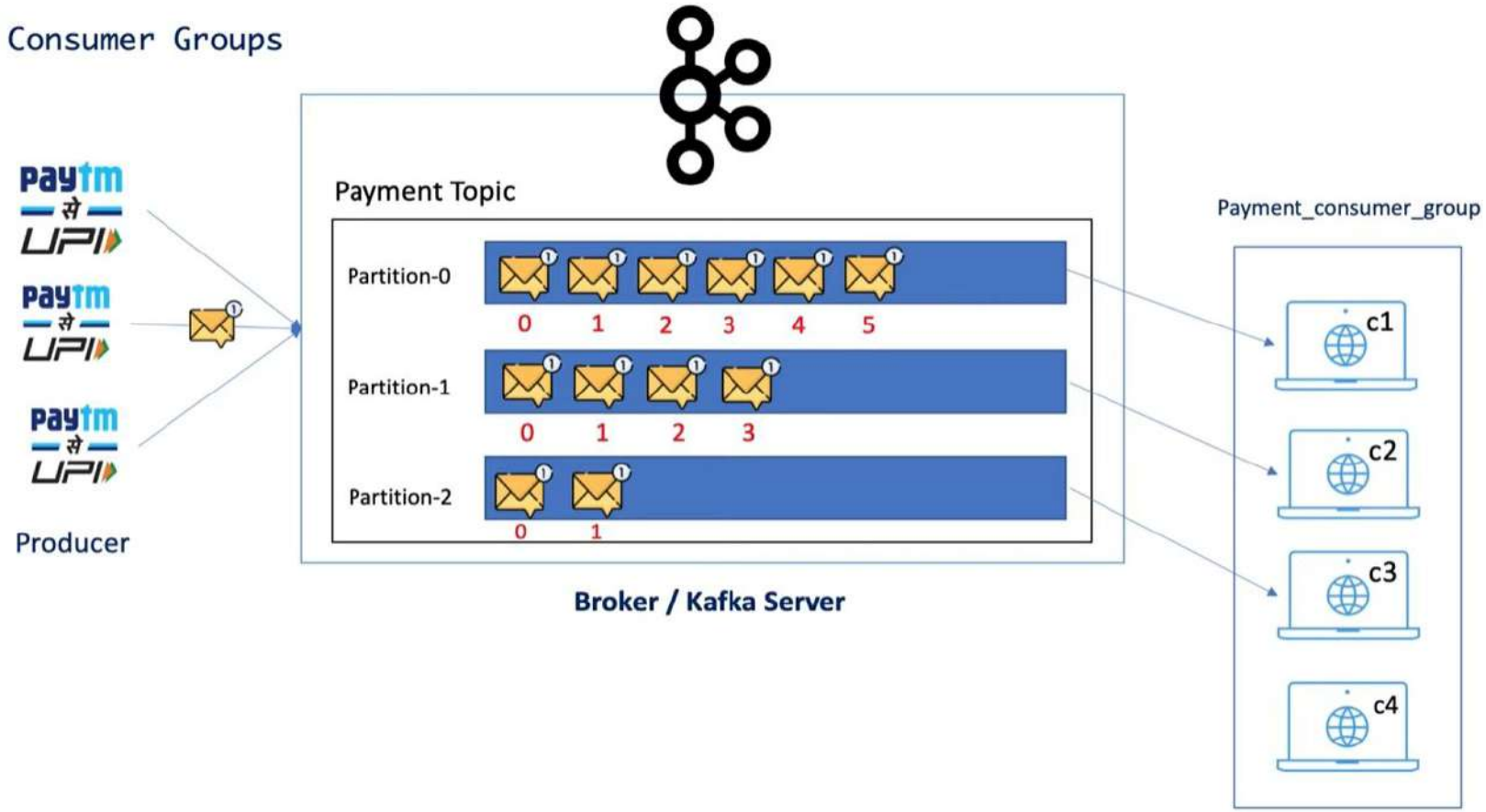

Paytm से UPI
Paytm से UPI
Paytm से UPI
PAYMENT-TOPIC
Partition-0
Partition-1
Partition-2
Broker / Kafka Server

Offset

In Kafka, a sequence number is assigned to each message in each partition of a Kafka topic. This sequence number is called Offset.



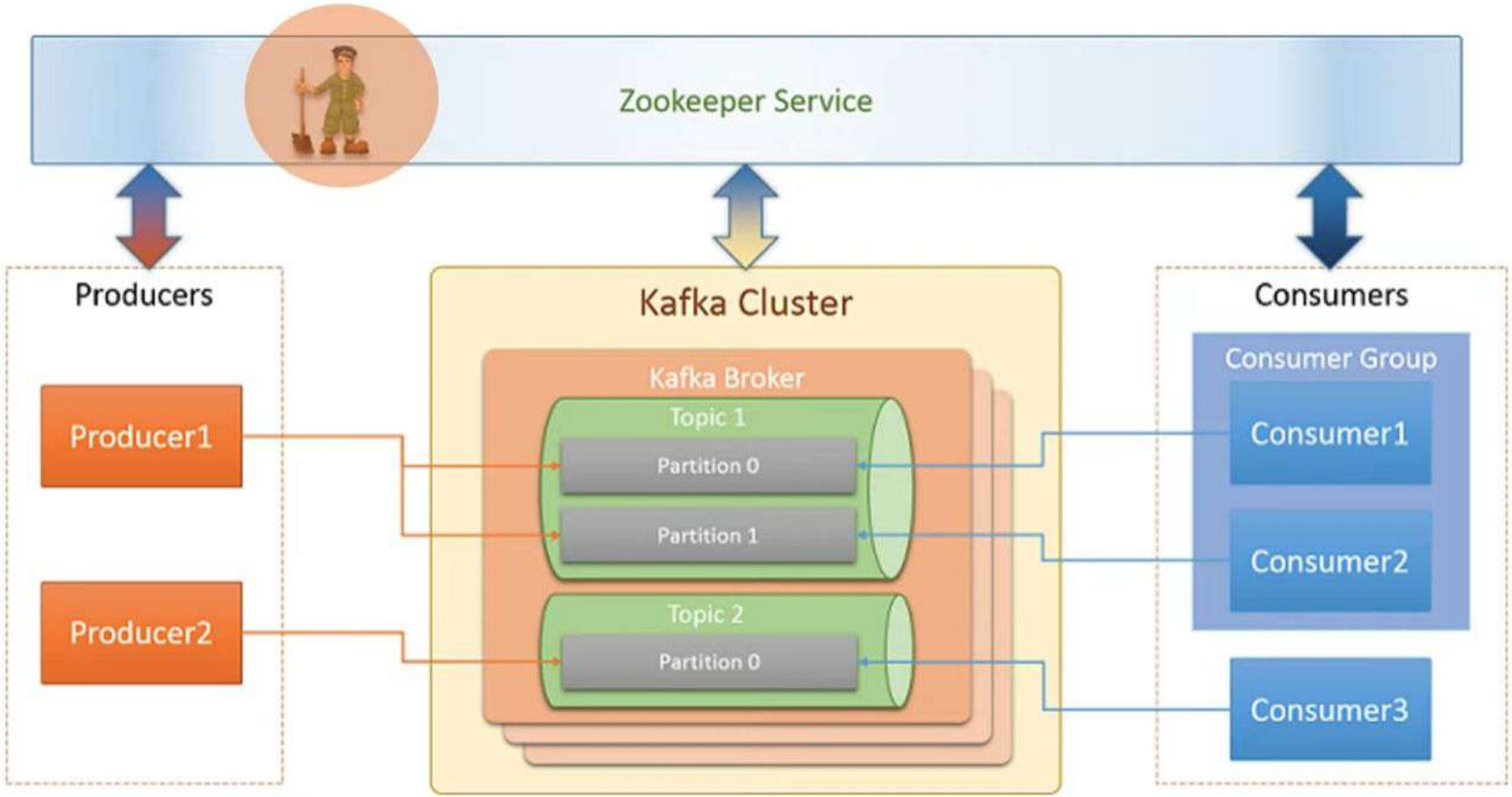
Consumer Groups



Zookeeper

Zookeeper is a prerequisite for Kafka. Kafka is a distributed system, and it uses Zookeeper for coordination and to track the status of Kafka cluster nodes. It also keeps track of Kafka topics, partitions, offsets, etc.

Zookeeper


Zookeeper Service
Producers
Producer1
Producer2
Kafka Cluster
Kafka Broker
Topic 1
Partition 0
Partition 1
Topic 2
Partition 0
Consumers
Consumer Group
Consumer1
Consumer2
Consumer3